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ELECTRODE ASSEMBLY AND ITS FORMING METHOD AND MANUFACTURING SYSTEM, SECONDARY BATTERY, BATTERY MODULE AND DEVICE

Abstract

The disclosure relates to an electrode assembly and its forming method and a manufacturing system, a secondary battery, a battery module and a device. The electrode assembly includes: a first electrode plate including a plurality of bending sections and a plurality of stacked first stacking sections, each bending section being configured to connect two adjacent first stacking sections, herein the bending section includes a guiding portion for guiding the bending section to be bent during production; and a second electrode plate with a polarity opposite to that of the first electrode plate, the second electrode plate including a plurality of second stacking sections, and each second stacking section being disposed between two adjacent first stacking sections.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/489,666, filed on Sep. 29, 2021, which is a continuation of International Application No. PCT/CN2020/081139, filed on Mar. 25, 2020, which claims priority to Chinese Patent Application No. 201911224967.7, filed on Dec. 4, 2019, all of which are incorporated herein by reference in their entireties.

FIELD
[0002] The disclosure relates to a technical field of battery, and in particular to an electrode assembly and its forming method and a manufacturing system, a secondary battery, a battery module and a device.
BACKGROUND
[0003] With developments of society, science and technology, secondary batteries are widely used to power high power devices, such as electric vehicles. The secondary battery forms a battery module by connecting a plurality of battery cells in series or in parallel to obtain a larger capacity or power.
[0004] The secondary battery includes a cathode electrode plate and an anode electrode plate, and the cathode electrode plate and the anode electrode plate are stacked to form an electrode assembly. However, after the cathode electrode plate and the anode electrode plate are stacked with respect to each other, at least one of the cathode electrode plate and the anode electrode plate may deviate from a predetermined position, which thereby affect an electrochemical performance of the secondary battery.
SUMMARY
[0005] Embodiments of the disclosure provide an electrode assembly and its forming method and a manufacturing system, a secondary battery, a battery module and a device, which can ensure that a first electrode plate and a second electrode plate are at predetermined positions after being stacked, thereby ensuring that the secondary battery has a good electrochemical performance.
[0006] In one aspect, according to embodiments of the disclosure, an electrode assembly for a secondary battery is provided, which includes: [0007] a first electrode plate including a plurality of bending sections and a plurality of stacked first stacking sections, each bending section being configured to connect two adjacent first stacking sections, wherein the bending section includes a guiding portion for guiding the bending section to be bent during production; and a second electrode plate with a polarity opposite to that of the first electrode plate, the second electrode plate including a plurality of second stacking sections, and each second stacking section being disposed between two adjacent first stacking sections.
[0008] In another aspect, according to embodiments of the disclosure, a secondary battery is provided, which includes an electrode assembly according to the above embodiments.

[0009] In another aspect, according to embodiments of the disclosure, a battery module is provided, which includes a secondary battery according to the above embodiments.
[0010] In another aspect, according to embodiments of the disclosure, a device is provided, which includes a secondary battery according to the above embodiments, and the secondary battery is configured to provide electrical energy.
[0011] In another aspect, according to embodiments of the disclosure, a forming method of an electrode assembly is provided, which includes: [0012] providing a first electrode plate, which includes a plurality of bending sections and a plurality of first stacking sections, each bending section being configured to connect two adjacent first stacking sections, wherein the bending section includes a guiding portion; [0013] providing a second electrode plate with a polarity opposite to that of the first electrode plate, wherein the second electrode plate includes a plurality of second stacking sections, and each of the second stacking sections is disposed between two adjacent first stacking sections; and [0014] bending the bending section along the guiding portion so that the two adjacent first stacking sections connected to the bending section are stacked.
[0015] In another aspect, an electrode assembly manufacturing system is provided according to embodiments of the disclosure, which includes: [0016] a first conveying mechanism for providing a first electrode plate, wherein the first electrode plate includes a plurality of bending sections and a plurality of first stacking sections, and each of the bending sections is configured to connect two adjacent first stacking sections; [0017] a trace making mechanism for providing a guiding portion in the bending section, wherein the guiding portion is configured to guide the bending section to be bent during production; [0018] a second conveying mechanism for providing a second electrode plate with a polarity opposite to that of the first electrode plate, wherein the second electrode plate includes a plurality of second stacking sections, and each of the second stacking sections is disposed between two adjacent first stacking sections; and [0019] a stacking mechanism for bending the bending section along the guiding portion and stacking the two adjacent first stacking sections connected to the bending section.
[0020] The disclosure provide following beneficial effects: by providing the guiding portion at the bending section, the bending section can be bent along the guiding portion during production, which ensures that the bending section has a more accurate bending position when it is bent with respect to the first stacking section, so that the first electrode plate and the second electrode plate are at predetermined positions after being stacked, thereby ensuring that the secondary battery has a good electrochemical performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] Features, advantages, and technical effects of exemplary embodiments of the disclosure will be described below with reference to accompanying drawings.
[0022] FIG. **1** is a schematic structural diagram of a vehicle disclosed in an embodiment of the disclosure;
[0023] FIG. **2** is a schematic diagram of an exploded structure of a battery pack disclosed in an embodiment of the disclosure;
[0024] FIG. **3** is a schematic diagram of a partial structure of a battery module disclosed in an embodiment of the disclosure;
[0025] FIG. **4** is a schematic diagram of an exploded structure of a secondary battery disclosed in an embodiment of the disclosure;
[0026] FIG. **5** is a partial top view structural diagram of a first electrode plate before being bent according to an embodiment of the disclosure;
[0027] FIG. **6** is a side view structural diagram of the first electrode plate of the embodiment as

shown in FIG. **5**;

[0028] FIG. **7** is a schematic structural view of the first electrode plate in the bent state of the embodiment as shown in FIG. **5**;

[0029] FIG. **8** is a schematic diagram of a connection structure of the first electrode plate of the embodiment as shown in FIG. **5**, a second electrode plate and a separator;

[0030] FIG. **9** is a side view structural diagram of an electrode assembly according to an embodiment of the disclosure;

[0031] FIG. **10** is a top view of a first structure diagram of the electrode assembly of the embodiment as shown in FIG. **9**;

[0032] FIG. **11** is a top view of a second structure diagram of the electrode assembly of the embodiment as shown in FIG. **9**;

[0033] FIG. **12** is a side view structural diagram of an electrode assembly according to another embodiment of the disclosure;

[0034] FIG. **13** is a schematic structural diagram of a first electrode plate in a bent state according to another embodiment of the disclosure;

[0035] FIG. **14** is a side view structural diagram of an electrode assembly including the first electrode plate of the embodiment as shown in FIG. **13**;

[0036] FIG. **15** is a schematic structural diagram of a first electrode plate in a bent state according to another embodiment of the disclosure;

[0037] FIG. **16** is a schematic structural diagram of a first electrode plate in a bent state according to another embodiment of the disclosure;

[0038] FIG. **17** is a cross-sectional structural diagram of a side view of an electrode assembly including the first electrode plate of the embodiment as shown in FIG. **16**;

[0039] FIG. **18** is a schematic structural diagram of a first electrode plate in a bent state according to another embodiment of the disclosure;

[0040] FIG. **19** is a side view structural diagram of an electrode assembly according to another embodiment of the disclosure;

[0041] FIG. **20** is a side view structural diagram of an electrode assembly according to another embodiment of the disclosure;

[0042] FIG. **21** is a schematic structural diagram of a stacked cell manufacturing system according to an embodiment of the disclosure;

[0043] FIG. **22** is a schematic structural diagram of a first conveying mechanism according to embodiments of the disclosure;

[0044] FIG. **23** is a schematic structural diagram of an anode electrode plate and a trace making mechanism according to embodiments of the disclosure;

[0045] FIG. **24** is a schematic diagram of an anode electrode plate with traces according to embodiments of the disclosure;

[0046] FIG. **25** is a schematic structural diagram of a second conveying mechanism according to embodiments of the disclosure;

[0047] FIG. **26** is a schematic structural diagram of a composite forming mechanism according to embodiments of the disclosure;

[0048] FIG. **27** is a schematic structural diagram of a heating conveying assembly according to embodiments of the disclosure;

[0049] FIG. **28** is a top view of the structure as shown in FIG. **27**;

[0050] FIG. **29** is a schematic structural diagram of a stacking mechanism according to embodiments of the disclosure;

[0051] FIG. **30** is a bottom view of the structure as shown in FIG. **29**;

[0052] FIG. **31** is a schematic diagram of a cooperation of a stacking mechanism, a main conveying mechanism and an assembly to be stacked according to embodiments of the disclosure.

[0053] The views are not necessarily plotted in actual proportion in the drawings.

REFERENCE SIGNS IN THE DRAWINGS

[0054] **1**. Vehicle; [0055] **10**. Battery Pack; [0056] **20**. Battery Module; [0057] **30**. Secondary Battery; [0058] **40**. Case; [0059] **50**. Electrode Assembly; **50***a*, Main Body; **50***b*, Tab; [0060] **51**. First Electrode Plate; **51***a*, Current Collector; **51***b*, Electrode Active Material Layer; **511**, Bending Section; **511***a*, Connecting Portion; **511***b*, Intermediate Transition Portion; **5110**, Thinned Area; **5111**, Guiding Portion; **5111***a*, Groove; **5111***b*, Through Hole; **5112**, Second Outer Edge; **512**, First Stacking Section; **5121**, First Outer Edge; [0061] **52**. Second Electrode Plate; **521**. Second Stacking Section; **5211**, Third Outer Edge; [0062] **53**. Separator; [0063] **60**. Top Cover Assembly; **61**. Top Cover Plate; **62**. Electrode Terminal; [0064] **70**. Connecting Plate; [0065] W. Extending Direction; H, Thickness Direction; X, First Direction; Y, Second Direction; Z, Bending Direction; [0066] **100**. First Conveying Mechanism; **101**. First Roll-Off Device; **102**. First Tape Splicing Device; **103**. First Tension Balancing Device; **104**. First Offset Correcting Device; **105**. First Dust Removing Device; [0067] **200**. Separator Conveying Mechanism; [0068] **300**. Trace Making Mechanism; **301**. First Trace Making Component; **302**. Second Trace Making Component; [0069] **400**. Second Conveying Mechanism; **401**. Second Roll-Off Device; **402**. Second Tape Splicing Device; **403**. Second Tension Balancing Device; **404**. Second Offset Correcting Device; **405**. Cutting Device; **406**. Second Dust Removing Device; **4061**, Belt Brush; **4062**, Dust Suction Device; [0070] **500**. Composite Forming Mechanism; **501**. Heating Conveying Assembly; **5011**, Heating Component; **5012**, Conveying Component; **5012***a*, Transmission Wheel; **5012***b*, Transmission Belt; **502**, Roller Pressing Component; **5021**, Squeezing Roller; **503**, Dust Removing Component; [0071] **600**, Stacking Mechanism; **601**, Power Source; **602**, Swing Mechanism; **6021**, Gap; [0072] **6022**, Mounting Base; **6023**, Clamping Roller; **6024**, Limiting Reinforcement Roller; **700**. Main Conveying Mechanism.

DETAILED DESCRIPTION

[0073] Below, embodiments of the disclosure will be further described in detail with reference to the drawings and embodiments. The detailed description according to the embodiments and the accompanying drawings are intended to exemplary illustrate the principles of the disclosure and are not intended to limit the scope of the disclosure, i.e., the disclosure is not limited to the described embodiments.

[0074] In the description of the disclosure, it should be noted that, unless otherwise stated, the meaning of “a plurality” is two or more; the orientation or positional relationship indicated by the terms “upper”, “lower”, “left”, “right”, “inner”, “outer” and the like is merely for the purpose of describing the disclosure and simplifying the description, and is not intended to indicate or imply that the device or component referred to has a particular orientation, is constructed and operated in a particular orientation, and therefore cannot be understood to be a limitation of the disclosure. Moreover, the terms “first”, “second”, and the like are configured for descriptive purposes only and are not to be construed as indicating or implying relative importance.

[0075] The orientations in the following description are all directions shown in the drawings and are not intended to limit specific structures of the disclosure. In the description of the disclosure, it should be noted that, unless otherwise stated, the terms “installation”, “connected to”, and “connected with” are to be understood broadly, and may be, for example, a fixed connection, a disassemble connection, or an integral connection; they can be connected directly or indirectly through an intermediate medium. The specific meaning of the above terms in the disclosure can be understood by the person skilled in the art according to actual circumstance.

[0076] For better understanding of the disclosure, embodiments of the disclosure will be described below in detail with reference to FIGS. **1** to **31**.

[0077] Embodiments of the disclosure provide a device that adopts a secondary battery **30** as a power source. The device can be but not limited to a vehicle, a ship, an aircraft or the like. Referring to FIG. **1**, an embodiment of the disclosure provides a vehicle **1** including a vehicle body and a battery module. The battery module is disposed in the vehicle body. Herein, the vehicle **1** may be a pure electric vehicle, a hybrid electric vehicle or an extended range vehicle. The vehicle body is provided with a drive motor electrically connected to the battery module. The battery module provides power to the drive motor. The drive motor is connected to wheels on the vehicle body through a transmission to drive the vehicle to travel. Optionally, the battery module may be horizontally disposed at a bottom of the vehicle body.

[0078] As shown in FIG. **2**, the battery module may be a battery pack **10**. There are various ways to construct the battery pack **10**. In some optional embodiments, the battery pack **10** includes a housing and a battery module **20** disposed in the housing. The number of the battery modules **20** may be one or more. The one or more battery modules **20** are disposed in the housing. The type of the housing is not limited. The housing can be a frame-shaped housing, a plate-shaped housing, a box-shaped housing or the like. Optionally, the housing includes a lower housing for accommodating the battery module **20** and an upper housing for closing the lower housing. The upper housing and the lower housing are closed to form an accommodating portion for accommodating the battery module **20**. It should be understood that the battery module may also be the battery module **20**, i.e., the battery module **20** is directly disposed on the vehicle body.

[0079] As shown in FIG. **3**, the battery module **20** includes a plurality of secondary batteries **30**. There are various ways to construct the battery module **20**. In one embodiment, the battery module **20** includes an accommodating portion and a plurality of secondary batteries **30** located in the accommodating portion. The secondary batteries **30** are arranged side by side in the accommodating portion. There are various ways to construct the accommodating portion: for example, the accommodating portion includes a casing and a cover plate for covering the casing; or the accommodating portion includes side plates and end plates that are successively connected in a closed way; or, the accommodating portion includes two ends plates opposite to each other and a strap surrounding the end plates and the secondary batteries **30**.

[0080] After noticing the problem of the poor electrochemical performance of existing secondary batteries, the inventors found that at least one of a cathode electrode plate and an anode electrode plate in a formed electrode assembly deviates from the predetermined position, which thus affects the electrochemical performance of the secondary battery. The inventors further discovered that at least one of the cathode electrode plate and the anode electrode plate in the formed electrode assembly deviates from the predetermined position, which results in lithium plating phenomenon in the electrode assembly and thereby affects the electrochemical performance of the secondary battery. It is speculated that the reason may be that a portion of the anode electrode plate extends beyond an outer edge of the cathode electrode plate but has a too small dimension or the anode electrode plate does not extend beyond the outer edge of the cathode electrode plate.

[0081] Through the analyzing an assembling process of the electrode assembly, the inventors further studied the lithium plating phenomenon and found that, by taking the anode electrode plate continuously arranged and the cathode electrode plate discontinuously arranged as an example, it is difficult for the anode electrode plate to be folded along a predetermined area during a bending process, leading to the small dimension of the portion of the anode electrode plate that extends beyond the outer edge of the cathode electrode plate after the cathode electrode plate and the anode electrode plate are stacked to form the electrode assembly, which will likely result in the lithium plating phenomenon in the electrode assembly and thereby affect the electrochemical performance and the safety performance of the secondary battery.

[0082] Based on the above problems discovered by the inventors, the inventors have improved a structure of an electrode assembly **50**, and embodiments of the disclosure will be further described below.

[0083] As shown in FIG. **4**, the secondary battery **30** according to an embodiment of the disclosure includes a case **40**, an electrode assembly **50** disposed in the case **40**, and a top cover assembly **60** sealingly connected to the case **40**.

[0084] The case **40** according to the embodiment of the disclosure has a square structure or other shapes. The case **40** includes an internal space for accommodating the electrode assembly **50** and electrolyte, and an opening communicating with the internal space. The case **40** may be made of materials such as aluminum, aluminum alloy, or plastic.

[0085] The top cover assembly **60** according to embodiments of the disclosure includes a top cover plate **61** and electrode terminals **62**. The top cover plate **61** according to embodiments of the disclosure includes an outer surface and an inner surface opposite to each other, and an electrode lead-out hole through the outer and inner surfaces. The top cover plate **61** can cover the opening of the case **40** and be sealingly connected to the case **40**. The inner surface of the top cover plate **61** faces the electrode assembly **50**. The electrode terminal **62** is disposed in the top cover plate **61** and disposed corresponding to the electrode lead-out hole. A part of the electrode terminal **62** is exposed from the outer surface of the top cover plate **61** and is configured to be welded with a busbar.

[0086] The electrode assembly **50** according to embodiments of the disclosure includes a main body **50***a* and tabs **50***b* extending from the main body **50***a*. In embodiments of the disclosure, from each of two opposite end faces of the main body **50***a*, one tab **50***b* extends. In some other embodiments, two tabs **50***b* extend from one of the two opposite end faces of the main body **50***a*. The two tabs **50***b* have opposite polarities, one of the two tabs is a negative tab and the other is a positive tab. A positive active material is coated on a coating region of the anode electrode plate, while a negative active material is coated on a coating region of the cathode electrode plate. A plurality of uncoated regions extending from the coating regions of the main body **50***a* are serve as tabs **50***b*. The positive tab extends from the coating region of the anode electrode plate while the negative tab extends from the coating region of the cathode electrode plate. Connecting plates **70** are configured to connect the tabs **50***b* of the electrode assembly **50** and the electrode terminals **62**.

[0087] In an embodiment, FIG. **5** schematically shows a partial structure of a first electrode plate **51** in an unfolded state. As shown in FIG. **5**, the first electrode plate **51** includes a plurality of bending sections **511** and a plurality of first stacking sections **512**, wherein the bending section **511** after being bent is at least partially in a bent state. The first electrode plate **51** has a continuously extending structure as a whole. Along an extending direction W of the first electrode plate **51**, the bending sections **511** and the first stacking sections **512** are alternately disposed. Each bending section **511** connects two adjacent first stacking sections **512**. Along a first direction X, each first stacking section **512** includes two opposite first outer edges **5121**, and each bending section **511** includes two opposite second outer edges **5112**. The first direction X is the same as a width direction of the first electrode plate **51**. The first direction X is perpendicular to the extending direction W. In the embodiment, along the first direction X, the first outer edges **5121** and the second outer edges **5112** on a same side are flush with each other. The first electrode plate **51** also includes tabs **50***b* extending beyond the first outer edges **5121** of the first stacking section **512** along the first direction X. The number and positions of the tabs **50***b* are arranged in one-to-one correspondence with the number and positions of the first stacking sections **512**. In the embodiment, the first electrode plate **51** includes a guiding portion **5111** disposed in the bending section **511**. The number of the guiding portions **5111** may be the same as the number of the bending sections **511**. It should also be understood that some bending sections **511** of all the bending sections **511** may be provided with the guiding portions **5111**, while the other bending sections **511** may not be provided with the guiding portions **5111**. The guiding portion **5111** is configured to guide the bending section **511** to be bent during production. In a production process, when an external force is applied to the first electrode plate **51** to perform a bending process, since the bending section **511** is provided with the guiding portion **5111**, it is easier for the bending section **511** to be bent at an area where the guiding portion **5111** is located, which accordingly facilitates improving a controllability and an accuracy of a bending position, thereby ensuring that the first electrode plate **51** and the second electrode plate **52** are located at respective predetermined positions so as to ensure that the secondary battery **30** has a good electrochemical performance.

[0088] FIG. **6** schematically shows a side view structure of the first electrode plate **51** according to the embodiment as shown in FIG. **5**. The first electrode plate **51** includes a current collector **51***a* and electrode active material layer **51***b* coated on the current collector **51***a*. The current collector **51***a* includes two surfaces opposite to each other in a thickness direction H of the first electrode plate **51**. Two electrode active material layers **51***b* are respectively disposed on the two surfaces. In an example, when the first electrode plate **51** is a cathode electrode plate, a material of the current collector **51***a* is a metal material such as aluminum or aluminum alloy. When the first electrode plate **51** is an anode electrode plate, a material of the current collector **51***a* is a metal material such as copper or copper alloy. The guiding portion **5111** may be a trace left by things. Optionally, it may refer to a structure formed by removing a part of the electrode active material layer **51***b* from the first electrode plate **51** through a material removing component, or a structure formed by removing a part of the electrode active material layer **51***b* and a part of the current collector **51***a* from the first electrode plate **51**.

[0089] In the embodiment, each bending section **511** is provided with a guiding portion **5111**. In the embodiment, the guiding portion **5111** includes a groove **5111***a*. The groove **5111***a* is recessed and extends from a surface of the first electrode plate **51** in the thickness direction H of the first electrode plate **51** along a direction approaching the current collector **51***a*. Among two adjacent bending sections **511**, a groove **5111***a* disposed in one bending section **511** is located at one side of the current collector **51***a*, and a groove **5111***a* disposed in the other bending section **511** is located at the other side of the current collector **51***a*. The groove **5111***a* may be formed by removing a part of the electrode active material layer **51***b* from the first electrode plate **51**. Alternatively, when the electrode active material is coated on the current collector **51***a*, less electrode active material is coated at a corresponding position to form the groove **5111***a*. In the embodiment, along the thickness direction H, a depth of the groove **5111***a* may be equal to a thickness of the electrode active material layer **51***b*. The groove **5111***a* extends to a surface of the current collector **51***a*, but does not extend into the current collector **51***a*. However, it should be understood that the depth of the groove **5111***a* may be less than the thickness of the electrode active material layer **51***b*, so that the groove **5111***a* does not pass through the electrode active material layer **51***b* in the thickness direction H; and in this case, there is a part of the electrode active material between the groove **5111***a* and the current collector **51***a*. In the embodiment, since the depth of the groove **5111***a* is less than or equal to the thickness of the electrode active material layer **51***b*, the current collector **51***a* will not be damaged when the groove **5111***a* is formed, and compared to a case where the depth of the groove **5111***a* is greater than the thickness of the electrode active material layer **51***b* (in this case, the current collector **51***a* will be damaged), a strength of the current collector **51***a* will not be affected. In the embodiment, a mouth of the groove **5111***a* is greater than or equal to a bottom of the groove **5111***a*. In an example, a projection of the groove **5111***a* on a plane perpendicular to the first direction X has a V-shape. However, the projection of the groove **5111***a* is not limited to a V shape, and may also have a U shape or a rectangular shape. Since the mouth of the groove **5111***a* is greater than or equal to the bottom of the groove **5111***a*, on one hand, it facilitates ensuring the bending position of the bending section **511** and enabling easy forming of the groove **5111***a*; and on the other hand, the electrode active material near the mouth of the groove **5111***a* is subjected to a small compression stress or no compression stress during the bending process, so that a bending resistance of the first stacking section **512** is reduced, and thereby the first stacking section **512** can be more easily and more accurately bent to a predetermined position. In the embodiment, the first electrode plate **51** further includes a thinned area **5110** disposed in the bending section **511**, and the thinned area **5110** is disposed corresponding to the groove **5111***a* along the thickness direction H of the first electrode plate **51**. A thickness of the first electrode plate **51** at the thinned area **5110** is smaller than a thickness of the first electrode plate **51** outside the thinned area **5110**. At this time, because a rigidity of the thinned area **5110** is less than a rigidity of the first electrode plate **51** outside the thinned area **5110**, the bending section **511** can be more easily bent at the thinned area

5110, which facilitates enabling first outer edges **5121** of two adjacent first stacking sections **512** on a same side to be aligned.

[0090] In the embodiment as shown in FIGS. **5** and **6**, in the extending direction W of the first electrode plate **51**, a thickness of the thinned area **5110** increases from a central area to two side areas. Along the first direction X, the groove **5111***a* extends to two opposite second outer edges **5112** of the bending section **511** to pass through the entire bending section **511**; and compared with a case where the groove **5111***a* does not pass through the entire bending section **511**, the electrode active material near the groove **5111***a* is subjected to a small compression stress or no compression stress during the bending process, so that the bending resistance of the first stacking section **512** is reduced, which thereby can better ensure the accuracy of the bending position of the bending section **511** and then ensure that the first stacking section **512** can be more easily and accurately bent to the predetermined position. In the embodiment, the bending section **511** includes a part of the electrode active material layer **51***b*. A part of one of the two electrode active material layers **51***b* corresponding to the thinned area **5110** is not completely removed, and a part of the other layer corresponding to the thinned area **5110** may be or may not be completely removed. In the embodiment, along the extending direction W of the first electrode plate **51**, an opening size of the groove **5111***a* is smaller than a size of the bending section **511**, so that other areas of the bending section **511** except the guiding portion **5111** are covered by the electrode active material layers **51***b*. In an example, both parts of the two electrode active material layers **51***b* corresponding to the thinned area **5110** are completely removed.

[0091] A size of the guiding portion **5111** in the first direction X is set according to a size of the bending section **511** in the first direction X. The size of the guiding portion **5111** in the first direction X is a length of the guiding portion **5111**. The size of the bending section **511** in the first direction X is also a length of the bending section **511**. Therefore, in some other embodiments, the groove **5111***a* does not pass through the bending section **511** along the first direction X. A ratio of the size of the groove **5111***a* in the first direction X to the size of the bending section **511** in the first direction X is from 0.4 to 0.8, preferably 0.4, 0.5, 0.6, 0.7 or 0.8.

[0092] FIG. **7** schematically shows a structure of the first electrode plate **51** according to the embodiment as shown in FIG. **5** in a multiple reciprocally folded state. During a manufacturing process of an electrode assembly **50**, a first electrode plate **51** should be bent. During production, the guiding portion **5111** can guide the bending section **511** to be bent, i.e., the bending section **511** can be bent along the guiding portion **5111**, so that the bending position can be located at a predetermined position, which facilitates ensuring first outer edges **5121** of two adjacent first stacking sections **512** on a same side to be aligned. The bending section **511** of the first electrode plate **51** is bent along a bending direction Z as shown in FIG. **7**. The bending direction Z and the first direction X are perpendicular to each other, i.e., a plane in which the bending direction Z is located is perpendicular to the first direction X. In the embodiment, the first electrode plate **51** can be bent reciprocally in a substantially zigzag shape. Dashed lines as shown in FIG. **7** do not indicate a physical structure, but schematically show separation lines between the bending sections **511** and the first stacking sections **512**. Since two adjacent grooves **5111***a* are located at two opposite surfaces of the first electrode plate **51**, and the groove **5111***a* is located at a side that bears a compression stress when the bending section **511** is being bent, after the first electrode plate **51** has been folded, an opening of the groove **5111***a* at the bending section **511** faces a space formed between two adjacent first stacking sections **512**, i.e., the groove **5111***a* is located at an inner surface of the bending section **511**, so that a side of the thinned area **5110** close to the groove **5111***a* does not bear a tensile stress, which reduces a possibility that the thinned area **5110** will be broken under the tensile stress. After the first electrode plate **51** has been folded, two adjacent first stacking sections **512** are stacked and spaced apart along the second direction Y, and the space formed between the two adjacent first stacking sections **512** is configured to accommodate the second stacking section **521** of the second electrode plate **52**. The second direction Y is the same as a stacking direction of the first stacking sections **512** and is perpendicular to the first direction X. In the embodiment, the bending section **511** after being bent has an arc shape, for example, it may have a circular arc shape.

[0093] In the embodiment shown in FIG. **8**, based on the first electrode plate **51**, in the thickness direction H of the first electrode plate **51**, separators **53** are respectively disposed on two opposite sides of the first electrode plate **51**. The two separators **53** are disposed in a pair, and the first electrode plate **51** is disposed between the two separators **53**. The separator **53** covers the first stacking section **512** and the bending section **511**. Along the first direction X as shown in FIG. **5**, at least some of the tabs **50***b* extend beyond an edge of the separator **53**. In the production process, the two separators **53** are attached to the first electrode plate **51** through a corresponding feeding equipment. A second electrode plate **52** is disposed on a side of the separator **53** away from the first electrode plate **51**. The first electrode plate **51** and the second electrode plate **52** have opposite polarities, and when one of them is a cathode electrode plate, the other is an anode electrode plate. The second electrode plate **52** includes a plurality of second stacking sections **521**. In the embodiment, two adjacent second stacking sections **521** are respectively disposed on opposite sides of the first electrode plate **51**. Along the thickness direction H, the first stacking section **512** and the second stacking section **521** are disposed corresponding to each other. In the embodiment, between two adjacent bending sections **511**, a second stacking section **521** is disposed. However, the disclosure is not limited to a case where a second stacking section **521** is disposed between two adjacent bending sections **511**, and may also provide a suitable number of second stacking sections **521** according to product requirements. In one example, after a separator **53** has been disposed on the first electrode plate **51**, the second stacking section **521** is attached to the separator **53**. For example, the second stacking section **521** and the separator **53** may be connected by hot pressing, electrophoresis, or bonding. The separator **53** is an insulator interposed between the first electrode plate **51** and the second electrode plate **52**. A material of the separator **53** may be an insulating material such as plastic to insulate and isolate the first electrode plate **51** from the second electrode plate **52**.

[0094] FIG. **9** schematically shows a side view structure of an electrode assembly **50** according to an embodiment. After the first electrode plate **51**, the separator **53**, and the second electrode plate **52** have been combined in a way as shown in FIG. **8**, the bending section **511** is bent under the guidance of the guiding portion **5111**, and finally a folded state as shown in FIG. **9** is formed. In the second direction Y, a second stacking section **521** is disposed between two adjacent first stacking sections **512**, so that the first stacking sections **512** and the second stacking section **521** are alternately arranged in sequence. In the second direction Y, the bending section **511** and the second stacking section **521** do not overlap each other. In the embodiment, the bending section **511** is completely in a bent state, and a starting line of the bending section **511** is an area where the bending section **511** starts to be bent with respect to the first stacking section **512**. There is an interval between the bending section **511** and the second stacking section **521**; and then both edges of the first stacking section **512** in the extending direction W extend beyond the second stacking section **521**, and an end of the second stacking section **521** facing the bending section **511** along the extending direction W is not in contact with the bending section **511**, which thereby reduces a possibility that the electrode active material drops in powder form or falls off due to an interference of the end of the second stacking section **521** with the bending section **511**. After the first electrode plate **51** has been bent, the groove **5111***a* at each bending section **511** is located on the inner surface of the bending section **511**, i.e., after the first electrode plate **51** has been bent, grooves **5111***a* at respective bending sections **511** are all located on a side of the current collector **51***a* close to the second stacking section **521**. Here, the inner surface refers to a surface of the bending section **511** close to the second stacking section **521**. Correspondingly, an outer surface of the bending section **511** refers to a surface of the bending section **511** away from the second stacking section **521**. In the embodiment, the guiding portion **5111** at the bending section **511** after being bent is disposed corresponding to a middle area of the second stacking section **521**.

[0095] FIG. **10** schematically shows a top view structure after the first stacking section **512** and the second stacking section **521** are stacked with respect to each other. In the embodiment, the first electrode plate **51** is an anode electrode plate, and the second electrode plate **52** is a cathode electrode plate. Under the guidance of the guiding portion **5111** at the bending section **511**, after the first electrode plate **51** is reciprocally folded, circumferential edges of the first stacking section **512** all extend beyond the second stacking section **521**, so as to ensure that the entire second stacking section **521** is covered by the first stacking sections **512**, which effectively reduces a possibility that lithium plating phenomenon is caused as the second stacking section **521** extends beyond the first stacking section **512**. Here, a case where the entire second stacking section **521** is covered by the first stacking section **512** means that an orthographic projection of the second stacking section **521** in the second direction Y is completely within an orthographic projection of the first stacking section **512** in the second direction Y, and then a projection area of the second stacking section **521** is smaller than a projection area of the first stacking section **512**. In an example, a distance between a first outer edge **5121** of the first stacking section **512** and a corresponding third outer edge **5211** of the second stacking section **521** is greater than or equal to 0.2 mm and less than or equal to 5 mm, preferably 0.5 mm, 1 mm, 1.5 mm, 2 mm, 2.5 mm, 3 mm, 3.5 mm, 4 mm or 4.5 mm. Since the bending section **511** includes the guiding portion **5111**, the first electrode plate **51** is bent under the guidance of the guiding portion **5111**, so that respective first outer edges **5121** on a same side of two adjacent first stacking sections **512** connected to the bending section **511** are aligned, i.e., two first outer edges **5121** of two adjacent first stacking sections **512** on the same side have an angle α. Here, a case where the first outer edges **5121** of the two adjacent first stacking sections **512** are aligned includes a state as shown in FIG. **10**, i.e., along the second direction Y, projections of the two adjacent first stacking sections **512** coincide with each other. The angle α between the two first outer edges **5121** of the two adjacent first stacking sections **512** on the same side is 0°, so that in a top view, the respective first outer edges **5121** of the two adjacent first stacking sections **512** are consistent with each other and are aligned. The case where the respective first outer edges **5121** of the two adjacent first stacking sections **512** are aligned may also include a state as shown in FIG. **11**. FIG. **11** schematically shows another top view structure after the first stacking section **512** and the second stacking section **521** have been stacked with respect to each other. The two first outer edges **5121** of the two adjacent first stacking sections **512** on the same side in the top view are not completely aligned. The two first outer edges **5121** of the two adjacent first stacking sections **512** on the same side have an angle α. The angle α is greater than 0° and less than or equal to 30° to ensure that the first stacking section **512** covers the second stacking section **521**. Preferably, the angle α is 5°, 10°, 15°, 20° or 25°. Here, the angle α is an angle with an allowable error. When a first outer edge **5121** of a first stacking section **512** after being bent is deviated so that it fails to coincide with a first outer edge **5121** of another first stacking section **512**, but it can still be ensured that the first stacking section **512** covers a second stacking section **521**, an angle between two first outer edges **5121** of two adjacent first stacking sections **512** on a same side is referred to as an angle with an allowable error.

[0096] For the first electrode plate **51** according to the embodiment of the disclosure, since the bending section **511** includes the guiding portion **5111**, when the first electrode plate **51** is being bent during the production process of the electrode assembly **50**, it is easier for the first electrode plate **51** to be bent at an area where the guiding portion **5111** of the bending section **511** is located under the guidance of the guiding portion **5111**, which accordingly facilitates improving a controllability and an accuracy of a bending position of the bending section **511** by providing the guiding portion **5111**, thereby improving an alignment of the first outer edges **5121** of the two adjacent first stacking sections **512**, reducing a possibility that one of the first stacking section **512** and the second stacking section **521** serving as a negative electrode cannot completely cover the other of the first stacking section **512** and the second stacking section **521** serving as a positive electrode due to a randomness of the bending position of the first electrode plate **51** after being bent, and reducing a possibility that lithium plating phenomenon occurs in the manufactured electrode assembly **50**. Furthermore, the electrode active material layer **51***b* coated on the current collector **51***a* has a certain brittleness. During the bending process of the bending section **511**, the electrode active material layer **51***b* will be subjected to an external force, so that the electrode active material layer **51***b* may fall off or drop in powder form from the current collector **51***a*, which affects an electrochemical performance and a safety performance of the electrode assembly **50**. The groove **5111***a* according to the disclosure is formed by reducing the corresponding electrode active material, so that during the bending process of the bending section **511**, the disposed groove **5111***a* facilitates reducing an internal stress suffered by the corresponding electrode active material layer **51***b*, thereby reducing a possibility that the electrode active material layer **51***b* will drop in powder form or fall off.

[0097] In some other embodiments, the same structure as the embodiments as shown in FIG. **7** and FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiments as shown in FIG. **7** and FIG. **9**. In the embodiment, a groove **5111***a* of one of two adjacent bending sections **511** is located on one surface of a first electrode plate **51**, and a groove **5111***a* of the other of the two adjacent bending sections **511** is located on the other surface of the first electrode plate **51**; and therefore, for the first electrode plate **51** after being bent, a guiding portion **5111** at the bending section **511** faces away from a space formed between two adjacent first stacking sections **512**, i.e., being located on an outer surface of the bending section **511**, so that the groove **5111***a* is located at a side of the bending section **511** that bears a tensile stress when the bending section **511** is being bent.

[0098] In some other embodiments, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, each bending section **511** is provided with a guiding portion **5111**. Grooves **5111***a* of respective guiding portions **5111** at two adjacent bending sections **511** are located on a same surface of the first electrode plate **51**, thus in the first electrode plate **51** after being bent, a groove **5111***a* at one of the two adjacent bending sections **511** is located on an outer surface of the bending section **511** so as to face away from a space formed between two adjacent first stacking sections **512**, and a groove **5111***a* at the other bending section **511** is located on an inner surface of the bending section **511** so as to face the space formed between the two adjacent first stacking sections **512**. In the embodiment, in a manufacturing process of the first electrode plate **51**, it is only needed to form respective grooves **5111***a* on a same surface of the first electrode plate **51**, which facilitates reducing a processing difficulty of the first electrode plate **51**.

[0099] FIG. **12** schematically shows a side view structure of an electrode assembly **50** according to another embodiment of the disclosure. In some other embodiments as shown in FIG. **12**, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, each bending section **511** is correspondingly provided with a guiding portion **5111**. One guiding portion **5111** includes two grooves **5111***a*. When the first electrode plate **51** is in an unfolded state, the two grooves **5111***a* are correspondingly disposed along the thickness direction H of the first electrode plate **51**. In the first electrode plate **51** after being bent, one of the two grooves **5111***a* is disposed on an outer surface of the bending section **511** so as to face away from a space formed between two adjacent first stacking sections **512**, and the other is disposed on an inner surface of the bending section **511** so as to face the space formed between the two adjacent first stacking sections **512**. In the embodiment, for each bending section **511**, the two grooves **5111***a* are disposed in the thickness direction H, and then the thinned area **5110** has a smaller thickness, which facilitates further reducing a rigidity of the thinned area **5110**. In this way, since the thinned area **5110** corresponding to the two grooves **5111***a* has a smaller rigidity, it is easier for the bending section **511** to be bent at an area where the thinned area **5110** is located, which facilitates further improving a controllability and an accuracy of the bending position. In this way, as compared to the bending section **511** having no groove **5111***a*, internal stresses of the electrode active material layers **51***b* on both sides according to the embodiment will be relatively small when being bent, which facilitates further reducing a bending difficulty and a possibility that the electrode active material layer **51***b* may fall off or drop in powder form from the current collector **51***a* due to tensile or compression stress. In one example, the two grooves **5111***a* have same structures.

[0100] In some other embodiments, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, the guiding portion **5111** includes a groove **5111***a*. The groove **5111***a* extends from an inner surface of the bending section **511** toward the current collector **51***a*. An opening of the groove **5111***a* faces the second stacking section **521**. The groove **5111***a* passes through the inner electrode active material layer **51***b* and a part of the current collector **51***a* along the thickness direction H of the first electrode plate **51**. In one example, at least one of a metal cutter, a laser cutter, and a liquid etching tool is adopted to form the groove **5111***a*. Therefore, in the embodiment, a part of the groove **5111***a* is allowed to be formed in the current collector **51***a*, so as to ensure that electrode active material in a corresponding area can be removed while processing accuracy requirements and a processing difficulty can also be reduced. Furthermore, since a part of the current collector **51***a* is removed to form a part of the groove **5111***a*, which thereby facilitates further reducing the rigidity of the thinned area **5110** disposed corresponding to the groove **5111***a* and enabling the bending section **511** to be more easily bent at the guiding portion **5111**. In other embodiments, the groove **5111***a* extends from an outer surface of the bending section **511** away from the second stacking section **521** toward the current collector **51***a*. The opening of the groove **5111***a* is located at the outer surface of the bending section **511**, so that the opening of the groove **5111***a* faces away from the second stacking section **521**. The groove **5111***a* passes through the outer electrode active material layer **51***b* and a part of the current collector **51***a* along the thickness direction H of the first electrode plate **51**.

[0101] FIG. **13** schematically shows a structure of a first electrode plate **51** according to another embodiment of the disclosure in a multiple reciprocally folded state. FIG. **14** schematically shows a side view structure of an electrode assembly **50** according to another embodiment of the disclosure. The first electrode plate **51** according to the embodiment as shown in FIG. **14** is the first electrode plate **51** according to the embodiment as shown in FIG. **13**. In the embodiments as shown in FIG. **13** and FIG. **14**, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, the guiding portion **5111** includes a groove **5111***a*. The groove **5111***a* is recessed and extends from an inner surface of the bending section **511** toward the current collector **51***a*. An opening of the groove **5111***a* faces the second stacking section **521**. The groove **5111***a* passes through the inner electrode active material layer **51***b* along the thickness direction H of the first electrode plate **51**, and along the extending direction W of the first electrode plate **51**, a size of the groove **5111***a* is equal to a size of the bending section **511**. A portion of the inner electrode active material layer **51***b* corresponding to the bending section **511** is completely removed so that a surface of the current collector **51***a* facing the second stacking section **521** is exposed. A portion of the outer electrode active material layer **51***b* corresponding to the bending section **511** is not removed but entirely remains. Since the inner electrode active material layer **51***b* at the bending section **511** is completely removed, the rigidity of the bending section **511** can be further reduced, so that it is easier for the bending section **511** to be bent at the guiding portion **5111**, and it also effectively avoids a case where the inner electrode active material layer **51***b* falls off or drops in powder form after the bending section **511** has been bent. In some other embodiments, the portion of the outer electrode active material layer **51***b* corresponding to the bending section **511** may be partially removed to form the guiding portion **5111** on the outer electrode active material layer **51***b*.

[0102] In some other embodiments, the same structure as the embodiment as shown in FIG. **14** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **14**. In the embodiment, the guiding portion **5111** includes a groove **5111***a*. The groove **5111***a* extends from the outer surface of the bending section **511** toward the current collector **51***a*. An opening of the groove **5111***a* faces away from the second stacking section **521**. The groove **5111***a* passes through the outer electrode active material layer **51***b* along the thickness direction H of the first electrode plate **51**, and along the extending direction W of the first electrode plate **51**, a size of the groove **5111***a* is equal to a size of the bending section **511**. The portion of the outer electrode active material layer **51***b* corresponding to the bending section **511** is completely removed, so that the surface of the current collector **51***a* facing away from the second stacking section **521** is exposed. The portion of the inner electrode active material layer **51***b* corresponding to the bending section **511** is not removed but entirely remains. In some other embodiments, the portion of the inner electrode active material layer **51***b* corresponding to the bending section **511** may also be partially removed to form the groove **5111***a* on the inner electrode active material layer **51***b*. Since the electrode active material layer **51***b* on the outer side of the bending section **511** is completely removed, the rigidity of the bending section **511** can be further reduced, so that it is easier for the bending section **511** to be bent at the guiding portion **5111**, and it also effectively avoids a case where the outer electrode active material layer **51***b* falls off or drops in powder form after the bending section **511** has been bent.

[0103] FIG. **15** schematically shows a structure of a first electrode plate **51** according to another embodiment of the disclosure in a multiple reciprocally folded state. In the embodiment as shown in FIG. **15**, the same structure as the embodiment as shown in FIG. **7** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **7**. In the embodiment, the guiding portion **5111** includes two or more grooves **5111***a*. Along the first direction X, the two or more grooves **5111***a* are spaced apart. In the embodiment, when the first electrode plate **51** is in an unfolded state, along the thickness direction H of the first electrode plate **51**, thinned areas **5110** are disposed corresponding to the grooves **5111***a*. The number and positions of the thinned areas **5110** are arranged in one-to-one correspondence with the number and positions of the grooves **5111***a*. In the embodiment, a projection of the groove **5111***a* on a plane perpendicular to the first direction X may have a triangle shape. However, the projection of the groove **5111***a* is not limited to a triangle shape, and may also have a trapezoid shape or a rectangular shape. Each groove **5111***a* extends from the outer surface of the bending section **511** toward the current collector **51***a*, so that the opening of the groove **5111***a* faces away from the second stacking section **521**. A ratio of a sum of sizes of respective grooves **5111***a* in the first direction X to a size of the bending section **511** in the first direction X is 0.4 to 0.8, preferably 0.4, 0.5, 0.6, 0.7 or 0.8. In some other embodiments, each groove **5111***a* extends from the inner surface of the bending section **511** toward the current collector **51***a*, so that the opening of the groove **5111***a* faces the second stacking section **521**. In some other embodiments, a plurality of grooves **5111***a* are disposed on the inner surface of the bending section **511**, and a plurality of grooves **5111***a* are disposed on the outer surface of the bending section **511**. In one example, when the first electrode plate **51** is in the unfolded state, along the thickness direction H of the first electrode plate **51**, the grooves **5111***a* on the inner surface and the grooves **5111***a* on the outer surface are located corresponding to each other. The positions of the thinned areas **5110**, the positions of the grooves **5111***a* on the inner surface and the positions of the groove **5111***a* on the outer surface are arranged in a one-to-one correspondence. A groove **5111***a* on the inner surface and a groove **5111***a* on the outer surface jointly and correspondingly provide a thinned area **5110**.

[0104] FIG. **16** schematically shows a structure of a first electrode plate **51** according to another embodiment of the disclosure in a multiple reciprocally folded state. In the embodiment as shown in FIG. **16**, the same structure as the embodiment as shown in FIG. **7** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **7**. In the embodiment, the guiding portion **5111** includes two or more through holes **5111***b*. Along the first direction X, two or more through holes **5111***b* are disposed at intervals. When the first electrode plate **51** is in an unfolded state, along the thickness direction H of the first electrode plate **51**, the through hole **5111***b* passes through the two electrode active material layers **51***b* and the current collector **51***a*. In the unfolded state of the first electrode plate **51**, a size of the through hole **5111** *b* in the extending direction W of the first electrode plate **51** is smaller than a size of the bending section **511** in the extending direction W of the first electrode plate **51**. In an example, a shape of the through hole **5111***b* may be a rectangular shape, a square shape, an ellipse shape, a trapezoid shape, or a triangle shape. In the embodiment, a ratio of a sum of sizes of respective through holes **5111***b* in the first direction X to a size of the bending section **511** in the first direction X is 0.4 to 0.8, preferably 0.6 or 0.7.

[0105] In one embodiment, the guiding portion **5111** includes a through hole **5111***b*. A ratio of a size of the through hole **5111***b* in the first direction X to a size of the bending section **511** in the first direction X is 0.4 to 0.8, preferably 0.6 or 0.7.

[0106] FIG. **17** schematically shows a side cross-sectional structure of an electrode assembly **50** according to another embodiment of the disclosure. The first electrode plate **51** included in the electrode assembly **50** is the first electrode plate **51** according to the embodiment as shown in FIG. **16**. In the embodiment, the through hole **5111***b* at the bending section **511** after being bent is disposed corresponding to a middle area of the second stacking section **521**. However, the disclosure does not limit the position of the through hole **5111***b*, and the position of the through hole **5111***b* can also be disposed corresponding to other areas of the second stacking section **521** that deviate from the middle area along the second direction Y.

[0107] FIG. **18** schematically shows a structure of a first electrode plate **51** according to another embodiment of the disclosure in a multiple reciprocally folded state. In the embodiment as shown in FIG. **18**, the same structure as the embodiment as shown in FIG. **7** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **7**. In the embodiment, the guiding portion **5111** includes two or more grooves **5111***a* and two or more through holes **5111***b*. Along the first direction X, one or more grooves **5111***a* may be disposed between two adjacent through holes **5111***b*. Alternatively, one or more through holes **5111***b* may be disposed between two adjacent grooves **5111***a*. In some other embodiments, the guiding portion **5111** may include any other number of grooves **5111***a* and any other number of through holes **5111***b* as needed. In an example, the guiding portion **5111** may include a groove **5111***a* and a through hole **5111***b*. As shown in FIG. **18**, at one bending section **511** of two adjacent bending sections **511**, each groove **5111***a* extends from an outer surface of the bending section **511** toward the current collector **51***a*, so that an opening of the groove **5111***a* faces away from the second stacking section **521**; and at the other bending section **511**, each groove **5111***a* extends from an inner surface of the bending section **511** toward the current collector **51***a*, so that an opening of the groove **5111***a* faces the second stacking section **521**. It should be understood that, in other embodiments, each groove **5111***a* at each bending section **511** extends from the outer surface of the bending section **511** toward the current collector **51***a*, so that the opening of the groove **5111***a* face away from the second stacking section **521**. Alternatively, each groove **5111***a* at each bending section **511** extends from the inner surface of the bending section **511** toward the current collector **51***a*, so that the opening of the groove **5111***a* faces the second stacking section **521**.

[0108] FIG. **19** schematically shows a side view structure of an electrode assembly **50** according to another embodiment of the disclosure. In the embodiment as shown in FIG. **19**, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, a bending section **511** includes two connecting portions **511***a* and an intermediate transition portion **511***b* connecting the two connecting portions **511***a*. The intermediate transition portion **511***b* is substantially perpendicular to the connecting portion **511***a*, and the intermediate transition portion **511***b* is substantially perpendicular to the first stacking section **512**. The two connecting portions

511*a* of the bending section **511** are respectively connected to two adjacent first stacking sections **512**. As shown in FIG. **19**, the connecting portion **511***a* is flush with the first stacking section **512**. Each bending section **511** is provided with two guiding portions **5111**. When the first electrode plate **51** is in an unfolded state, the two guiding portions **5111** are disposed at an interval along the extending direction W of the first electrode plate **51**. A part of each guiding portion **5111** is located at the connecting portion **511***a*, and the other part is located at the intermediate transition portion **511***b*. In the embodiment, both two guiding portions **5111** are disposed on an inner surface of the bending section **511**. An inner side of the intermediate transition portion **511***b* includes a part of an electrode active material layer **51***b*. In the embodiment, when the first electrode plate **51** is being bent, it can be easily bent at positions corresponding to the two guiding portions **5111**, which thereby further improves a controllability and an accuracy of the bending position, and further ensures that first outer edges **5121** of the two adjacent first stacking sections **512** are aligned.
[0109] FIG. **20** schematically shows a side view structure of an electrode assembly **50** according to another embodiment of the disclosure. In the embodiment as shown in FIG. **20**, the same structure as the embodiment as shown in FIG. **9** will not be repeated here, and the description here will mainly focus on differences from the embodiment as shown in FIG. **9**. In the embodiment, the separator **53** extends beyond the first electrode plate **51**, and a portion of the separator **53** that extends beyond the first electrode plate **51** wraps around the first electrode plate **51** and the second electrode plate **52**, so that the separator **53** directly forms an insulation protection for the first electrode plate **51** and the second electrode plate **52**, which reduces subsequent steps of insulating and packaging the first electrode plate **51** and the second electrode plate **52** that have been folded.
[0110] The electrode assembly **50** according to embodiments of the disclosure includes a first electrode plate **51**, a second electrode plate **52** and a separator **53**. The first electrode plate **51** includes a first stacking section **512** and a bending section **511** that are alternately disposed. The bending section **511** includes a guiding portion **5111**. During the production process of the electrode assembly **50**, it is required to sequentially dispose the separator **53** and the second electrode plate **52** on the first electrode plate **51**, and then bend the first electrode plate **51** through multiple reciprocal folding, so that the first stacking section **512** and the second stacking section **521** of the second electrode plate **52** are stacked with respect to each other. The guiding portion **5111** of the bending section **511** can guide the first electrode plate **51** to bend at a predetermined position of the bending section **511** during the bending process of the first electrode plate **51**, which thereby improves a controllability and an accuracy of the bending position of the first electrode plate **51**, and then ensures first outer edges **5121** of the first stacking sections **512** to be aligned, so that one of the first stacking section **512** and the second stacking section **521** serving as a cathode electrode plate can cover the other of the first stacking section **512** and the second stacking section **521** serving as an anode electrode plate. In this way, the electrode assembly **50** according to embodiments of the disclosure has a low possibility that lithium plating phenomenon occurs between the first electrode plate **51** and the second electrode plate **52**, and ensures that a secondary battery using the electrode assembly **50** has a good electrochemical performance and a good safety performance.
[0111] Embodiments of the disclosure also provide a forming method of an electrode assembly **50**, which includes: [0112] providing a first electrode plate **51**, the first electrode plate **51** including a plurality of bending sections **511** and a plurality of first stacking sections **512**, and each bending section **511** being configured to connect two adjacent first stacking sections **512**, wherein the bending section **511** includes a guiding portion **5111**; [0113] providing a second electrode plate **52** with a polarity opposite to that of the first electrode plate **51**, the second electrode plate **52** including a plurality of second stacking sections **521**, and each second stacking section **521** being disposed between two adjacent first stacking sections **512**; [0114] bending the bending section **511** along the guiding portion **5111** so that first outer edges **5121** of the two adjacent first stacking sections **512** connected to the bending section **511** are aligned.

[0115] The forming method of the electrode assembly **50** according to embodiments of the disclosure can be configured to manufacture the electrode assembly **50** according to the above embodiments.
[0116] In an embodiment, the forming method further includes a step of forming the guiding portion **5111** through at least one of a metal cutter, a laser cutter, and a liquid etching tool. In an example, in this step, the electrode active material layer **51***b* at a predetermined position on the bending section **511** is removed by mechanical cutting, laser cutting, water erosion, or chemical reaction, etc., so as to form the guiding portion **5111** at the first electrode plate **51**.
[0117] In one embodiment, prior to the step of providing the second electrode plate **52** with the polarity opposite to that of the first electrode plate **51**, the first electrode plate **51** including the guiding portion **5111** is provided with separators **53** disposed in pairs, so that the separators **53** disposed in pairs are located on opposite sides of the first electrode plate **51**.
[0118] In the forming method of the electrode assembly **50** according to the disclosure, during a production process of the electrode assembly **50**, the first electrode plate **51** is bent along the guiding portion **5111** of the bending section **511**. The guiding portion **5111** of the bending section **511** can guide the first electrode plate **51** to be bent at a predetermined position of the bending section **511** during the bending process of the first electrode plate **51**, thereby improving controllability and accuracy of the bending position of the first electrode plate **51**, so that the first outer edges **5121** of the two adjacent first stacking sections **512** connected to the bending section **511** can be aligned, and one of the first stacking section **512** and the second stacking section **521** serving as a cathode electrode plate can cover the other of the first stacking section **512** and the second stacking section **521** serving as an anode electrode plate. In this way, the electrode assembly **50** manufactured by the forming method of the electrode assembly **50** according to embodiments of the disclosure has a low possibility that lithium plating phenomenon occurs between the first electrode plate **51** and the second electrode plate **52**, and ensures that a secondary battery using the electrode assembly **50** has a good electrochemical performance and a good safety performance.
[0119] In embodiments of the disclosure, the electrode assembly **50** may be a stacked cell formed by stacking and folding a first electrode plate **51**, a separator **53**, and a second electrode plate **52**. The first electrode plate **51** includes a plurality of bending sections **511** and a plurality of first stacking sections **512**, wherein the bending section **511** after being bent is at least partially in a bent state. The first electrode plate **51** has a continuously extending overall structure. Along an extending direction W of the first electrode plate **51**, the bending sections **511** and the first stacking sections **512** are alternately disposed. The second electrode plate **52** includes a plurality of second stacking sections **521**, and each second stacking section **521** is disposed between two adjacent first stacking sections **512**. In the following embodiments, description is made by taking an example in which first electrode plate **51** is an anode electrode plate and the second electrode plate **52** is a cathode electrode plate. Similarly, in other embodiments, the first electrode plate **51** may be a cathode electrode plate, and the second electrode plate **52** may be an anode electrode plate. The guiding portion **5111** may be a trace formed in the first electrode plate **51**.
[0120] In order to better understand the disclosure, a stacked cell manufacturing system and a stacked cell forming method according to embodiments of the disclosure will be described in detail below with reference to FIGS. **21** to **32**.
[0121] Embodiments of the disclosure provide a stacked cell manufacturing system, including:
[0122] a first conveying mechanism **100** for providing an anode electrode plate, wherein the anode electrode plate includes a plurality of bending sections **511** and a plurality of first stacking sections **512**, and each bending section **511** is configured to connect two adjacent first stacking sections **512**; [0123] a trace making mechanism **300**, wherein the trace making mechanism **300** is configured to provide traces in the bending section **511**, and the traces are configured to guide the bending section **511** to bend during a production process; [0124] a second conveying mechanism **400** for providing a cathode electrode plate with a polarity opposite to that of the anode electrode

plate, wherein the cathode electrode plate includes a plurality of second stacking sections **521**, and each second stacking section **521** is disposed between two adjacent first stacking sections **512**;

[0125] a stacking mechanism **600** for bending the bending section **511** along the trace and stacking the two adjacent first stacking sections **512** connected to the bending section **511**.

[0126] In one embodiment, please refer to FIG. **21**, the stacked cell manufacturing system according to embodiments of the disclosure includes a first conveying mechanism **100**, a separator conveying mechanism **200**, a trace making mechanism **300**, a second conveying mechanism **400**, and a composite forming mechanism **500** and a stacking mechanism **600**. The first conveying mechanism **100** is configured to provide an anode electrode plate. The separator conveying mechanism **200** is arranged downstream of the first conveying mechanism **100** and is configured to provide separators **53** disposed in pairs, and the separators **53** disposed in pairs are configured to clamp the anode electrode plate. The trace making mechanism **300** is arranged upstream of the separator conveying mechanism **200**. In some optional examples, the trace making mechanism **300** may be located between the first conveying mechanism **100** and the separator conveying mechanism **200**, and the trace making mechanism **300** is configured to provide traces in the anode electrode plate. The second conveying mechanism **400** is arranged downstream of the separator conveying mechanism **200** and is configured to provide a plurality of cathode electrode plates to the separator **53**. The composite forming mechanism **500** is arranged downstream of the second conveying mechanism **400** and is configured to combine the anode electrode plate, the separator **53** and the cathode electrode plate to form an assembly to be stacked. The stacking mechanism **600** is arranged downstream of the composite forming mechanism **500**, and the stacking mechanism **600** is configured to reciprocally fold and stack the assemblies to be folded and stacked along the traces to form a stacked cell.

[0127] It should be noted that the terms “upstream” and “downstream” as mentioned above and below in the disclosure refer to orders in a production sequence of the stacked cell, and is not intended to limit spatial positions of the respective components.

[0128] Moreover, the traces as mentioned above and below in the disclosure refer to marks or imprints left by things, such as a crease, and optionally, it may refer to a structure formed by a material-removed portion after a part of material is removed from the anode electrode plate through a material removing component.

[0129] The stacked cell manufacturing system according to embodiments of the disclosure can meet production requirements of the stacked cell, and also can reduce safety risks of the stacked cell.

[0130] Please also refer to FIG. **22**, optionally, the first conveying mechanism **100** may include a first roll-off device **101**, a first tape splicing device **102**, a first tension balancing device **103** and a first offset correcting device **104**. The first offset correcting device **104** is arranged downstream of the first roll-off device **101**, and both the first tape splicing device **102** and the first tension balancing device **103** are located between the first roll-off device **101** and the first offset correcting device **104**.

[0131] The first roll-off device **101** may include a first roll-off roller and a driving component that drives the first roll-off roller to rotate, the anode electrode plate is wound around the first roll-off roller, and the first roll-off roller rotates to roll off the anode electrode plate.

[0132] Optionally, the first tape splicing device **102** can be arranged downstream of the first roll-off device **101**, and can be used to perform tape splicing when the rolling off of the anode electrode plate has been finished to ensure a continuous production.

[0133] Optionally, the first offset correcting device **104** is located upstream of the trace making mechanism **300**; it can be monitored, through a detection device, in real time or at a certain time interval whether the anode electrode plate is within a predetermined range of the trace making mechanism **300**; if not, it is required to adjust the position of the anode electrode plate to ensure that the anode electrode plate is always within a trace making range of the trace making mechanism

300.

[0134] Optionally, the first tension balancing device **103** may be located downstream of the first tape splicing device **102**; when the first roll-off device **101** and a driving motor for providing a moving power for the anode electrode plate are not synchronized, the first tension balancing device **103** may be used to adjust it to maintain a tension of the anode electrode plate within a certain range.

[0135] Please refer to FIG. **23** and FIG. **25**, in some optional embodiments, the trace making mechanism **300** includes a first trace making component **301** and a second trace making component **302** arranged at an interval, and in a thickness direction H of the anode electrode plate, the trace making component **301** is configured to provide traces on one surface of the anode electrode plate, and the second trace making component **302** is configured to provide traces on the other surface of the anode electrode plate. Through the above arrangement, the trace making mechanism **300** in operation can alternately provide traces on the two surfaces of the anode electrode plate in the thickness direction H, so that the anode electrode plate can be easily folded along the traces on the two surfaces of the anode electrode plate during a final stacking process. A direction of the trace is consistent with that of a final crease, which facilitates enabling the stacking mechanism **600** to reciprocally fold and stack the assemblies to be stacked along the traces.

[0136] As an optional embodiment, the first trace making component **301** is one of a metal cutter, a laser cutter, and a liquid etching tool, the first trace making component **301** in the above form can remove material at a predetermined position by mechanical cutting, laser cutting, water erosion, or chemical reaction, etc., to form a trace on one surface of the anode electrode plate in the thickness direction H. The operation process is simple, and facilitates an easy formation of traces.

[0137] In some optional examples, similarly, the second trace making component **302** of the stacked cell manufacturing system according to the above embodiments is one of a metal cutter, a laser cutter, and a liquid etching tool. The second trace making component **302** can remove material at a predetermined position by mechanical cutting, laser cutting, water erosion, or chemical reaction, etc., to form a trace on the other surface of the anode electrode plate in the thickness direction H. The operation process is simple, and facilitates an easy formation of traces.

[0138] As an optional embodiment, a first dust removing device **105** is disposed downstream of the trace making mechanism **300**. The first dust removing device **105** is located between the trace making mechanism **300** and the separator conveying mechanism **200**. The first dust removing device **105** is configured to remove dust from a front side and/or a back side of the anode electrode plate to clean the anode electrode plate. The first dust removing device **105** may include a bristle brush and a dust suction component. When the anode electrode plate is moving, the dust can be peeled off by the bristle brush, and the dust peeled off from the anode electrode plate can be sucked and collected by the dust suction component to ensure a cleanliness of the anode electrode plate when it is clamped by the separator **53** and thereby further optimize an electrical performance of the stacked cell formed by folding and stacking.

[0139] In some optional embodiments, the separator conveying mechanism **200** may be further located downstream of the second trace making component **302** of the trace making mechanism **300**, the separator conveying mechanism **200** includes separator conveying devices disposed in pairs, and two separator conveying devices in a pair can be disposed opposite to each other. Each separator conveying device includes a separator roll-off roller **21** and a driving component that drives the separator roll-off roller to rotate, the separator **53** is wound around the separator roll-off roller, and the separator roll-off roller rotates to roll off the separator **53**, and a guiding wheel may guide the corresponding separator **53** to a predetermined position to clamp the anode electrode plate with traces.

[0140] Please also refer to FIG. **25**, in some optional embodiments, the second conveying mechanism **400** may include a second roll-off device **401**, a second tape splicing device **402**, a second tension balancing device **403**, a second offset correcting device **404**, a cutting device **405**

and a second dust removing device **406**.
[0141] Optionally, the second roll-off device **401** may include a second roll-off roller and a driving component that drives the second roll-off roller to rotate, the cathode electrode plate is wound around the second roll-off roller, and the second roll-off roller rotates to roll off the cathode electrode plate.
[0142] Optionally, the second tape splicing device **402** can be arranged downstream of the second roll-off device **401**, and can be used to perform tape splicing when the rolling off of the cathode electrode plate has been finished to ensure a continuous production.
[0143] Optionally, the second tension balancing device **403** is located downstream of the second tape splicing device **402**; when the second roll-off device **401** and a driving motor for providing a moving power for the cathode electrode plate are not synchronized, the second tension balancing device can be used **403** may be used to adjust it to maintain a tension of the cathode electrode plate within a certain range. Optionally, the second offset correcting device **404** is located downstream of the second tension balancing device **403**; it can be monitored, through a detection device, in real time or at a certain time interval whether the cathode electrode plate is within a predetermined range of the second offset correcting device **404**; if not, it is required to adjust the position of the cathode electrode plate to ensure that the cathode electrode plate is always within a cutting range of the cutting mechanism **45**.
[0144] Optionally, the cutting device **405** is disposed downstream of the second offset correcting device **404**, and is configured to cut the tape-shaped cathode electrode plate into a plurality of block-shaped structures with a predetermined size.
[0145] Optionally, a second dust removing device **406** is located downstream of the cutting device **405**, and is configured to receive the block-shaped cathode electrode plate and remove dust from the cathode electrode plate, so as to ensure a cleanliness of the cathode electrode plate connected to the separator **53**. The second dust removing device **406** may include a belt brush **4061** and a dust suction device **4062**; the cathode electrode plate cut by the cutting device **405** falls to the belt brush **4061** of the second dust removing device **406**, and the cathode electrode plate can be transported through the belt brush **4061** in a direction towards the composite forming mechanism **500** and connected to the separator **53**. In the transportation process, the dust on the cathode electrode plate can also be peeled off by the belt brush **4061** and sucked and collected by the dust suction device **4062**, so as to ensure a cleanliness of the cathode electrode plate connected to the separator **53** and thereby enable the produced stacked cell to better meet its electrical requirements.
[0146] In specific implementations, the second conveying mechanisms **400** may be disposed in pairs according to requirements, and the second conveying mechanisms **400** disposed in pairs may be opposite to each other, and may synchronously or alternately provide cathode electrode plates to a same separator **53** or different separators **53**.
[0147] Please refer to FIGS. **26-28**, optionally, the composite forming mechanism **500** of the stacked cell manufacturing system according to the above embodiments may include a heating conveying assembly **501** and a roller pressing component **502**; the heating conveying assembly **501** is configured to heat and convey the separator **53** and the cathode electrode plate; and the roller pressing component **502** is disposed downstream of the heating conveying assembly **501** and is configured to roll the separator **53** and the cathode electrode plate after being heated, which then are compositely connected. The composite forming mechanism **500** adopts the above structure form, has a simple structure, and can ensure a composite effect of the cathode electrode plate and the separator **53** that clamps the anode electrode plate.
[0148] As an optional embodiment, the heating conveying assembly **501** includes a heating component **5011** and a conveying component **5012**. The heating component **5011** is configured to heat the separator **53** and the cathode electrode plate. The conveying component **5012** includes a transmission wheel **5012***a* and a transmission belt **5012***b* cooperating with the transmission wheel **5012***a*, and the transmission belt **5012***b* is disposed around the heating component **5011** and is

configured to convey the separator **53** and the cathode electrode plate.
[0149] Since the cathode electrode plate is applied with a PVDF adhesive and the corresponding separator **53** is also applied with a PVDF adhesive, both adhesives after being heated and pressed can be better bonded together.
[0150] Therefore, the heating conveying assembly **501** adopts the above structures, which can meet heating and bonding requirements, and also the conveying component **5012** is restricted to include a transmission wheel **5012***a* and a transmission belt **5012***b* that cooperates with the transmission wheel **5012***a*; by restricting a relationship between the transmission belt **5012***b* and the heating component **5011**, on the basis that heating requirements are met, the cathode electrode plate on the surface of the separator **53** can be protected and transported by the transmission belt **5012***b*, so that the cathode electrode plate can move synchronously with the separator **53**, to ensure a stable relative position of the cathode electrode plate and the separator **53**, thereby ensuring composite requirements between the cathode electrode plate and the separator **53**.
[0151] Moreover, in the stacked cell manufacturing system according to embodiments of the disclosure, the heating conveying assembly **501** adopts the above structural forms, so that after the cathode electrode plate has been sufficiently heated by the heating component **5011**, the cathode electrode plate and the separator **53** are connected together through the roller pressing component **502** to achieve a preparation of the assembly to be stacked. Compared with a traditional composite forming mechanism, the heating conveying assembly **501** adopts the transmission belt to replace a disposable PET film; and by cancelling the PET film, it can save time taken to unwind and wind the PET film, and can increase a utilization rate of equipment and reduce a manufacturing cost.
[0152] In some optional embodiments, the heating component **5011** may adopt components capable of providing thermal energy such as an oven, a heat exchanger, etc., to at least complete heating of the cathode electrode plate and the separator **53**.
[0153] In some optional embodiments, when the heating component **5011** adopts an oven structure, the oven can be made of a metal plate and include multiple heating tubes evenly disposed within the oven; heat from the heating tubes can bring a heating oven to a setting temperature; when the cathode electrode plate passes through the oven, the temperature of the oven makes the cathode electrode plate and the separator **53** reach a certain temperature by heat radiation.
[0154] Optionally, the transmission belt **5012***b* can be a belt, the number of which can be arranged according to a size of the cathode electrode plate; in some optional embodiments, the number of the transmission belts **5012***b* can be two or more, and the two or more transmission belts **5012***b* are disposed at intervals and can jointly convey the cathode electrode plate and the separator **53**, to ensure a stability of a force applied to the cathode electrode plate, thereby ensuring that the cathode electrode plate can smoothly move in synchronization with the separator **53**.
[0155] As an optional embodiment, the number of heating conveying assemblies **501** is two or more, every two heating conveying assemblies **501** form a group and are disposed opposite to each other, and the two heating conveying assemblies **501** in a same group jointly clamp and convey the cathode electrode plate and the separator **53** through oppositely disposed transmission belts **5012***b*.
[0156] By disposing the heating conveying assemblies **501** in pairs and enabling the heating conveying assemblies **501** disposed in pairs to jointly clamp and act on the cathode electrode plate and the separator **53**, it can be ensured that the cathode electrode plates on two separators **53** can move in synchronization with respective corresponding separators **53**, which can better ensure a stability of a relative position of each cathode electrode plate and each separator **53** so as to ensure an accuracy of a position of the cathode electrode plate on the separator **53** before the roller pressing component **502**.
[0157] As an optional embodiment, the roller pressing component **502** may include squeezing rollers **5021** disposed in pairs, and the cathode electrode plate and the separator **53** after being heated can be squeezed by the squeezing rollers **5021** disposed in pairs, so that the cathode electrode plate and the separator **53** are compositely connected and form an assembly to be stacked

with the anode electrode plate.
[0158] As an optional embodiment, the roller pressing component **502** and/or the transmission belt **5012***b* is provided with a dust removing component **503**, i.e., at least one of the roller pressing component **502** and the transmission belt **5012***b* is provided with a dust removing component **503**, and the dust removing component **503** can also adopt a cooperation of a bristle brush and a dust suction device to remove dust from the assembly to be stacked to better ensure a performance of the stacked cell.
[0159] Please refer to FIGS. **29** to **31**, in some optional embodiments, in the stacked cell manufacturing system according to the above embodiments, the stacking mechanism **600** may include a power source **601** and a swing mechanism **602**; the swing mechanism **602** includes a gap **6021** through which the assembly to be stacked can pass; and the power source **601** is connected to the swing mechanism **602** and drives the swing mechanism **602** to swing back and forth along a predetermined trajectory to reciprocally fold and stack the assembly to be stacked to form a stacked cell. The stacking mechanism **600** adopts the above structural form, has a simple structure and a low cost, and can reciprocally fold the assembly to be stacked according to the traces on the anode electrode plate, so that the formed stacked cell has a better performance.
[0160] As an optional embodiment, the swing mechanism **602** includes a mounting base **6022** and clamping rollers **6023** disposed in pairs and connected to the mounting base **6022**, the gap **6021** is formed between the clamping rollers **6023** disposed in pairs, and the swing mechanism **602** is connected to the power source **601** through the mounting base **6022**. The swing mechanism **602** adopts the above structural form, which allows it to be easily connected to the power source **601** to better meet power transmission requirements, and also to meet passing through requirements of the assembly to be stacked, so that the assembly to be stacked can be folded back and forth along the predetermined traces to ensure packing requirements of the stacked cell.
[0161] Optionally, the power source **601** may adopt a driving motor, the mounting base **6022** may include mounting plates disposed in pairs and at intervals, the clamping roller **6023** is located between two mounting plates and axial ends of the clamping roller **6023** are respectively connected to the corresponding mounting plates.
[0162] In some optional embodiments, the swing mechanism **602** further includes limiting reinforcement rollers **6024** disposed in pairs, and the limiting reinforcement roller **6024** is located upstream of the clamping roller **6023** and connected to the mounting base **6022**. By providing the limiting reinforcement roller **6024**, a tilt angle of the assembly to be stacked can be limited when the swing mechanism **602** is moving, which better ensures that the assembly to be stacked can be folded back and forth along the corresponding traces, thereby ensuring a stacking accuracy.
[0163] Moreover, by providing the limiting reinforcement roller **6024**, it can also strengthen the mounting base **6022**, can avoid changes in a relative position of the mounting base **6022** and the clamping roller **6023** or a relative position of the clamping rollers **6023** disposed in pairs due to the swinging of the swing mechanism **602** along the predetermined trajectory, and can also better ensure the stacking requirements of the stacked cell.
[0164] In some optional embodiments, the stacked cell manufacturing system according to the above embodiments further includes a main conveying mechanism **700**, the main conveying mechanism **700** is located between the composite forming mechanism **500** and the stacking mechanism **600**, and the main conveying mechanism **700** is configured to provide a moving power to the assembly to be stacked to better ensure that the assembly to be stacked moves toward the stacking mechanism **600** at a predetermined speed. The driving motor configured to provide power to the anode electrode plate described in the above embodiments may be the main conveying mechanism **700**.
[0165] Therefore, the stacked cell manufacturing system according to embodiments of the disclosure includes a first conveying mechanism **100**, a separator conveying mechanism **200**, a trace making mechanism **300**, a second conveying mechanism **400**, a composite forming

mechanism **500**, and a stacking mechanism **600**; and by providing the trace making mechanism **300** and enabling it to make the traces on the anode electrode plate, a step of cutting the anode electrode plate can be omitted, which avoids burr generation, ensures safety of the stacked cell, and also simplifies a structure of the stacked cell manufacturing system.
[0166] In one embodiment, please refer to FIGS. **21** to **32**, embodiments of the disclosure also provide a forming method of a stacked cell, including: [0167] S**100**: providing an anode electrode plate, and making multiple traces on the anode electrode plate, wherein the multiple traces are distributed at intervals in an extending direction W of the anode electrode plate; [0168] S**200**: providing separators **53** disposed in pairs to the anode electrode plate with the traces, and enabling the separators **53** disposed in pairs to jointly clamp the anode electrode plate; [0169] S**300**: providing multiple cathode electrode plates to the separator **53**, so that the multiple cathode electrode plates are applied at intervals along the extending direction W and connected to a surface of the separator **53** away from the anode electrode plate to form an assembly to be stacked, wherein each cathode electrode plate is located between two adjacent traces; [0170] S**400**, reciprocally folding and stacking the assembly to be stacked along positions of the multiple traces to form a stacked cell.
[0171] In some optional examples, the forming method of the stacked cell according to the embodiments of the disclosure may be implemented through the stacked cell manufacturing system discussed in the above embodiments.
[0172] In step S**100**, the provided anode electrode plate has a continuous tape-like structure, and the trace is a structure formed by a material-removed portion after a part of material is removed from the anode electrode plate by a material removing component, wherein the material removing component is one of a metal cutter, a laser cutter, and a liquid etching tool, i.e., the material removing component may be the trace making mechanism **300** described in the above embodiments.
[0173] In some optional examples, the part of the material removed by the material removing component from the anode electrode plate may be an electrode active material, and then a depth of the trace is less than or equal to a thickness of the electrode active material layer **51***b*. The part of the material removed by the material removing component from the anode electrode plate may also be the electrode active material and a material of the current collector **51***a*, and then the depth of the trace is greater than the thickness of the electrode active material layer **51***b*.
[0174] In some optional examples, in step S**100**, one trace of the two adjacent traces is located on one surface of the anode electrode plate in its thickness direction H, and the other trace is located on the other surface of the anode electrode plate in the thickness direction H.
[0175] When the forming method of the disclosure is implemented through the stacked cell manufacturing system according to any of the above embodiments, in step S**100**, the anode electrode plate may be provided by the first conveying mechanism **100**, and the corresponding traces may be made on the anode electrode plate by the trace making mechanism **300**. In step S**200**, the separators **53** disposed in pairs may be provided by the separator transport mechanism **200**.
[0176] In some optional embodiments, in step S**300**, one of two adjacent cathode electrode plates is connected to one of the separators **53** disposed in pairs, and the other is connected to the other of the separators **53** disposed in pairs. Through the above arrangement, the formed stacked cell can better meet use requirements and optimize an electrical performance of the stacked cell.
[0177] In some optional examples, the cathode electrode plate may be provided by the second conveying mechanism **400** in the stacked cell manufacturing system according to any of the above embodiments.
[0178] In step S**400**, the stacking mechanism **600** in the stacked cell manufacturing system according to any of the above embodiments may fold and stack the assembly to be stacked to meet production requirements of the stacked cell.
[0179] The forming method of the stacked cell according to the embodiments of the disclosure can

meet the production requirements of the stacked cell, and can also reduce safety risks of the stacked cell.

[0180] It should be noted that, in the stacked cell manufacturing system and the forming method of the stacked cell according to the above embodiments of the disclosure, in the thickness direction H of the anode electrode plate, a thickness of an area of the anode electrode plat where the trace is formed is less than a thickness of other area of the anode electrode plate where no trace is formed. Making the traces can ensure that it is easier for the anode electrode plate to be folded at the area where the trace is formed as compared to the other areas.

[0181] Optionally, the trace may be the groove **5111***a* according to the above embodiments which is formed after a material is removed from the anode electrode plate. Optionally, the shape of the groove **5111***a* may be a U-shaped groove, a triangular groove, or other regular-shaped polygonal grooves or irregularly shaped special-shaped grooves. Optionally, the trace passes through the anode electrode plate in a tape width direction of the anode electrode plate. The tape width direction of the anode electrode plate is the same as the first direction X, and the tape width direction is perpendicular to the extending direction W and the thickness direction H.

[0182] Optionally, the number of grooves **5111***a* is two or more. Along the first direction X, two or more grooves **5111***a* are disposed at intervals. Alternatively, the number of grooves **5111***a* may be one.

[0183] Optionally, the trace may be the through hole **5111***b* according to the above embodiments which is formed after a material is removed from the anode electrode plate. When the first electrode plate **51** is in an unfolded state, along the thickness direction H of the first electrode plate **51**, the through hole **5111***b* passes through the two electrode active material layers **51***b* and the current collector **51***a*. In an example, a shape of the through hole **5111***b* may be a rectangle shape, a square shape, an ellipse shape, a trapezoid shape, or a triangle shape. The number of the through holes **5111***b* is two or more, and along the first direction X, the two or more through holes **5111***b* are disposed at intervals. In one embodiment, the number of through holes **5111***b* may be one.

[0184] Optionally, the traces may be the groove **5111***a* and the through hole **5111***b* according to the above embodiments which are formed after the material is removed from the anode electrode plate. Optionally, the number of the through holes **5111***b* is two or more, and along the first direction X, one or two or more grooves **5111***a* may be disposed between two adjacent through holes **5111***b*. Alternatively, the number of grooves **5111***a* is two or more, and one or more through holes **5111***b* may be disposed between two adjacent grooves **5111***a*.

[0185] Although the disclosure has been described with reference to the preferred embodiments, various modifications may be made to the disclosure and components may be replaced with equivalents without departing from the scope of the disclosure; and in particular, the technical features mentioned in the various embodiments can be combined in any manner as long as there is no structural conflict. The disclosure is not limited to the specific embodiments disclosed herein, but includes all technical solutions falling within the scope of the claims.

Claims

1. An electrode assembly manufacturing system, wherein the electrode assembly manufacturing system comprises: a first conveying mechanism for providing a first electrode plate, wherein the first electrode plate comprises a plurality of bending sections and a plurality of first stacking sections, and each of the bending sections is configured to connect two adjacent first stacking sections; a trace making mechanism for providing a guiding portion in the bending section, wherein the guiding portion is configured to guide the bending section to be bent during production; a second conveying mechanism for providing a second electrode plate with a polarity opposite to that of the first electrode plate, wherein the second electrode plate comprises a plurality of second stacking sections, and each of the second stacking sections is disposed between two adjacent first stacking sections; and a stacking mechanism for bending the bending section along the guiding portion and stacking the two adjacent first stacking sections connected to the bending section.

2. The electrode assembly manufacturing system according to claim 1, wherein the trace making mechanism comprises a first trace making component and a second trace making component disposed at an interval, the first trace making component is configured to provide the guiding portion on one surface of the first electrode plate, and the second trace making component is configured to provide the guiding portion on the other surface of the first electrode plate.

3. The electrode assembly manufacturing system according to claim 2, wherein the first trace making component is one of a metal cutter, a laser cutter, and a liquid etching tool; and/or the second trace making component is one of a metal cutter, a laser cutter, and a liquid etching tool.

4. The electrode assembly manufacturing system according to claim 1, wherein the electrode assembly manufacturing system further comprises a separator conveying mechanism; the separator conveying mechanism is located downstream of the first conveying mechanism and upstream of the second conveying mechanism, the separator conveying mechanism is configured to provide separators disposed in pairs, and the separators disposed in pairs are configured to clamp the first electrode plate.

5. The electrode assembly manufacturing system according to claim 4, wherein the electrode assembly manufacturing system further comprises a composite forming mechanism for combining the first electrode plate, the separator and the second electrode plate to form an assembly to be stacked; the composite forming mechanism comprises: a heating conveying assembly for heating and conveying the separator and the second electrode plate; and a roller pressing component which is disposed downstream of the heating conveying assembly and is configured to roll the separator and the second electrode plate after being heated to compositely connect the separator and the second electrode plate.

6. The electrode assembly manufacturing system according to claim 5, wherein the heating conveying assembly comprises: a heating component for heating the separator and the second electrode plate; and a conveying component, which comprises a transmission wheel and a transmission belt cooperating with the transmission wheel, wherein the transmission belt is disposed around the heating component and is configured to convey the separator and the second electrode plate.

7. The electrode assembly manufacturing system according to claim 6, wherein the number of the heating conveying assemblies is two or more, every two heating conveying assemblies form a group and are disposed opposite to each other, and the two heating conveying assemblies in a same group jointly clamp and convey the second electrode plate and the separator through oppositely disposed transmission belts.

8. The electrode assembly manufacturing system according to claim 6, wherein the roller pressing component and/or the transmission belt is provided with a dust removing component.

9. The electrode assembly manufacturing system according to claim 1, wherein the stacking mechanism comprises a power source and a swing mechanism, and the power source is connected to the swing mechanism and drives the swing mechanism to swing back and forth along a predetermined trajectory to reciprocally fold and stack the assembly to be stacked to form an electrode assembly.

10. The electrode assembly manufacturing system according to claim 9, wherein the swing mechanism comprises a mounting base and clamping rollers disposed in pairs and connected to the mounting base, the swing mechanism comprises a gap through which the assembly to be stacked can pass, the gap is formed between the clamping rollers disposed in pairs, and the swing mechanism is connected to the power source through the mounting base.

11. The electrode assembly manufacturing system according to claim 10, wherein the swing mechanism further comprises limiting reinforcement rollers disposed in pairs, and the limiting reinforcement roller is located upstream of the clamping roller and connected to the mounting base.

12. The electrode assembly manufacturing system according to claim 5, wherein the electrode assembly manufacturing system further comprises a main conveying mechanism, the main conveying mechanism is located between the composite forming mechanism and the stacking mechanism, and the main conveying mechanism is configured to provide a moving power to the assembly to be stacked.